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AI-BASED LEGAL DOCUMENT MANAGEMENT SYSTEM

Software Engineering Problem Analysis Report

1. Understand the Problem Statement

1.1 Problem in Our Own Words

Law firms manage many legal documents, making manual searching and tracking difficult. The proposed AI-Based Legal Document Management System securely stores and organizes documents in one place. It uses AI to classify documents and find information quickly. OCR extracts text from scanned files. The system also manages legal cases, tracks document versions, sends deadline reminders, and protects confidential information. It reduces manual work, saves time, and improves efficiency.

1.2 Objectives

- Store and organize legal documents digitally in one place.
- Use AI to classify documents and search information quickly.
- Improve security, track document versions, and manage deadlines.
- Extract text from scanned documents using OCR.
- Reduce manual work and improve document management efficiency.

1.3 Scope of the System

The system is intended for lawyers, legal assistants, law firms, and in-house legal departments. Its scope includes user login, role-based permissions, document upload and storage, optical character recognition (OCR) for scanned files, AI-assisted classification and information extraction, intelligent search, version history, case management, reminders, audit logs, and basic reports. AI suggestions should be reviewed by authorized users. The system supports document administration; it does not replace professional legal judgement or provide guaranteed legal advice.

2. Identify Key Stakeholders and Challenges

2.1 Stakeholders, Roles, and Responsibilities

- **Lawyers:** Review, manage, and access legal documents related to cases. They use the system to search for important information quickly.
- **Legal Assistants:** Upload, organize, and update legal documents. They also track important dates and ensure documents are available when needed.
- **Managers:** Monitor legal cases, check work progress, and review reports to ensure that documents and tasks are managed properly.
- **System Administrators:** Manage user accounts, assign access permissions, and maintain system security to prevent unauthorized access.
- **Clients:** Provide necessary legal documents and information to lawyers for handling their cases.
- **IT Team:** Develop, maintain, and update the system. They fix technical problems and ensure that the system works efficiently..

2.2 Major Challenges

- **Document Management:** Handling large numbers of legal documents.
- **Searching:** Finding documents quickly.
- **AI Accuracy:** Incorrect document classification or text extraction.
- **Data Security:** Preventing unauthorized access.
- **Version Control:** Managing different document versions.
- **Deadline Management:** Avoiding missed legal deadlines.
- **Maintenance:** Managing system updates and costs.

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3. Analyze the Current Approach

3.1 Existing System or Process

The existing system mainly uses manual methods to store, organize, and search legal documents. Deadlines are tracked using calendars or spreadsheets, while document classification and version management are handled manually. This process is time-consuming and may cause errors, security issues, and missed deadlines.

3.2 Strengths of the Current Approach

- Paper files and familiar folders are easy for staff to understand.
- Small teams may begin using manual methods with little initial setup.
- Users can directly inspect the original document and apply professional judgement.
- Existing office software may already be available to the organization.

3.3 Limitations and Gaps

The existing system is time-consuming and makes document searching difficult. Manual classification may cause errors, and managing multiple versions can create confusion. It also lacks intelligent search, automated reminders, and centralized security, increasing the risk of missed deadlines and unauthorized access.

4. Proposed Software-Based Solution

4.1 System Overview

The AI-Based Legal Document Management System provides a centralized platform to store, organize, and manage legal documents securely. It uses AI to classify documents, OCR to extract text, and intelligent search to find files quickly. The system also supports version control, case management, deadline reminders, and secure user access..

4.2 Major Functional Modules

1. **User and Access Management**: registration or administrator-created accounts, secure login, user profiles, and role-based permissions.
2. **Document Repository and OCR**: upload, store, preview, download, organize, and extract text from supported scanned documents.
3. **AI Classification and Intelligent Search**: recommend categories, extract key information, and search by title, keyword, category, or available document text.

4. **Case Management and Deadline Reminders**: create cases, associate documents with cases, track case status, and configure reminders for important dates and obligations.
5. **Security, Version Control, and Reporting**: preserve document versions, record access and changes, manage security settings, and produce basic activity and document reports.

4.3 Functional Requirements

- **User Login:** Allow users to log in securely.
- **Document Upload:** Upload, view, and download legal documents.
- **OCR:** Extract text from scanned documents.
- **AI Classification:** Automatically categorize documents.
- **Search:** Find documents using keywords and categories.
- **Case Management:** Create and manage legal cases.
- **Deadline Reminders:** Send reminders for important dates.
- **Version Control:** Track document changes and previous versions.
- **Security:** Control access to confidential documents.
- **Reports:** Generate document and activity reports.

4.4 Non-Functional Requirements

- **Security:** Protect confidential legal documents.
- **Usability:** Make the system easy to use.
- **Reliability:** Prevent data loss and ensure proper functioning.
- **Performance:** Provide fast document searches.
- **Scalability:** Support increasing users and documents.

4.5 Application of Software Engineering Principles

- **Modularity:** Divide the system into separate modules.
- **Security:** Protect confidential legal documents.
- **Scalability:** Support more users and documents in the future.
- **Reliability:** Prevent data loss and system failures.
- **Maintainability:** Make the system easy to update.
- **Usability:** Provide a simple, user-friendly interface.

5. Justify the Chosen Methodology/Framework

5.1 Selected Methodology: Agile Scrum

Agile Scrum is a project management method that divides work into small tasks called user stories and completes them in short periods called sprints.

5.2 How Scrum Will Be Applied

- **Product Backlog:** List all project requirements and user stories.
- **Sprint Planning:** Select tasks for a short sprint.
- **Development and Testing:** Develop and test the selected features.
- **Sprint Review:** Demonstrate completed work and collect feedback.
- **Retrospective:** Identify improvements for the next sprint.
- **Backlog Refinement:** Update and prioritize upcoming tasks.

5.3 Support for Development and Maintenance

- **Easy Development:** Develop the system module by module.
- **Regular Testing:** Identify and fix errors early.
- **User Feedback:** Improve features based on user suggestions.
- **Easy Updates:** Add new features and security improvements when needed.
- **Bug Fixing:** Correct problems to maintain system performance.

6. Expected Outcomes and Limitations

6.1 Expected Benefits

- **Faster Searching:** Find legal documents quickly.
- **Better Organization:** Store documents in one place.
- **Reduced Manual Work:** Use AI to classify documents.
- **Deadline Tracking:** Receive reminders for important dates.
- **Improved Security:** Protect confidential information.
- **Easy Tracking:** Maintain document versions and activity records.

6.2 How the Solution Addresses the Challenges

The system organizes legal documents, makes searching faster, protects confidential information, sends deadline reminders, and tracks document changes to reduce errors and improve efficiency.

6.3 Risks, Implementation Challenges, and Limitations

- **AI Errors**: AI may classify documents incorrectly.
- **OCR Limitations**: Poor-quality scanned documents may reduce text extraction accuracy.
- **Security Risks**: Confidential information may be exposed.
- **Migration**: Transferring existing documents may take time.
- **Training**: Users may need training to use the system.
- **Maintenance Costs**: Regular updates and storage require resources.

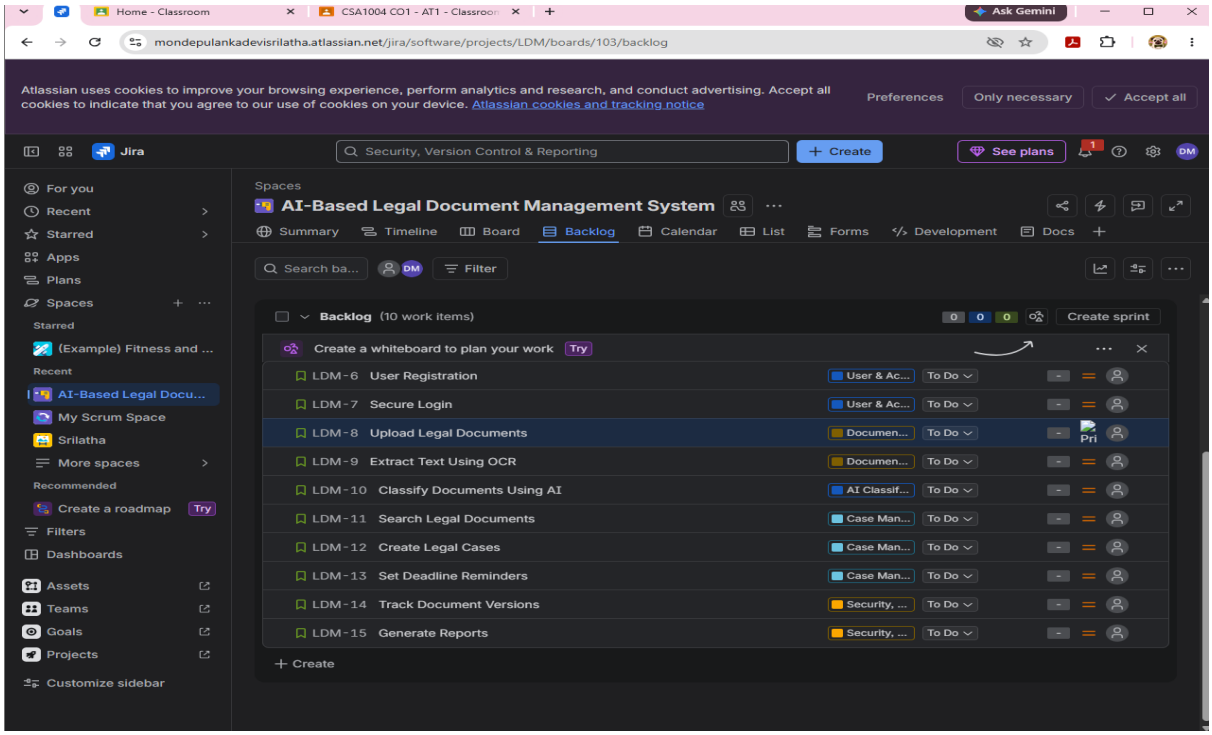
6.4 Future Enhancements

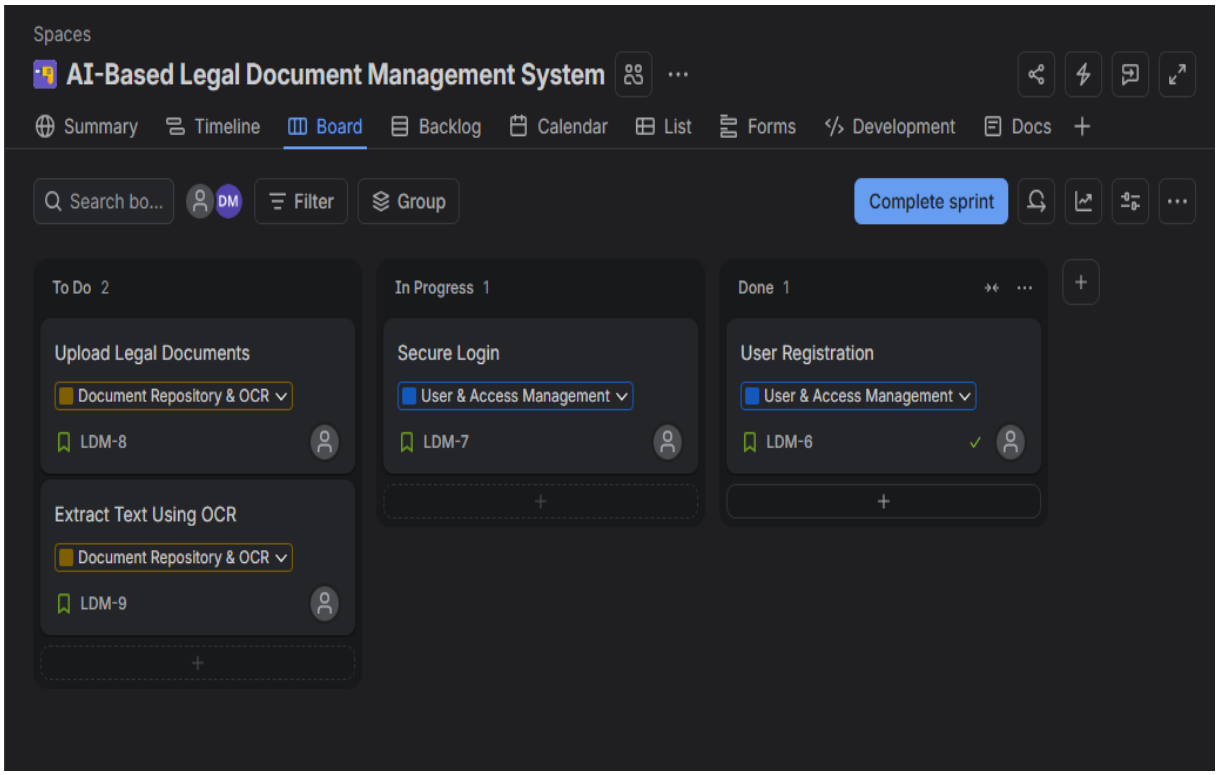
- **Advanced AI:** Improve document classification and accuracy.
- **Voice Search:** Search documents using voice commands.
- **Multilingual Support:** Support multiple languages.
- **Cloud Integration:** Access documents from anywhere.
- **Advanced Security:** Add biometric authentication and encryption.
- **Automated Legal Analysis:** Identify important clauses and summarize legal documents.
- **Mobile Application:** Manage legal documents through smartphones.

Conclusion

The AI-Based Legal Document Management System proposes a secure and organized way to store, classify, search, and track legal documents. By combining modular software design, AI assistance, access controls, version history, reminders, and audit trails, it can reduce routine administrative effort and improve document retrieval. Agile Scrum supports incremental delivery and stakeholder feedback. Successful implementation will depend on data quality, security, user training, testing, and careful human review of AI-generated information.

SCREENSHOTS / OUTPUT EVIDENCE





Submitted By

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